SPY Strategy Study: the six original strategies and SMA10/30

All strategies use the unified engine: signal at the close, execution at the next open, 10 bps per side, lagged 3-month T-bill on cash, close-to-close marking, and band rebalancing of partial positions.

Replication of the original notebook (same method: open-to-open returns, 0% cash, rf=0 Sharpe)

Period	Approach	Metric	Notebook	This lab	Diff
Development 70%	SMA 20/50 net	CAGR	0.085	0.085	+0.000
Development 70%	SMA 20/50 net	Sharpe (rf=0)	0.780	0.780	+0.000
Development 70%	SMA 20/50 net	Max drawdown	-0.243	-0.243	+0.000
Development 70%	Buy-and-hold	CAGR	0.144	0.144	+0.000
Development 70%	Buy-and-hold	Sharpe (rf=0)	0.920	0.920	-0.000
Development 70%	Buy-and-hold	Max drawdown	-0.320	-0.320	-0.000
Holdout 30%	SMA 20/50 net	CAGR	0.038	0.038	+0.000
Holdout 30%	SMA 20/50 net	Sharpe (rf=0)	0.381	0.381	-0.000
Holdout 30%	SMA 20/50 net	Max drawdown	-0.283	-0.283	-0.000
Holdout 30%	Buy-and-hold	CAGR	0.136	0.136	+0.000
Holdout 30%	Buy-and-hold	Sharpe (rf=0)	0.826	0.826	-0.000
Holdout 30%	Buy-and-hold	Max drawdown	-0.263	-0.263	+0.000

Holdout start: notebook 2021-09-29, lab 2021-09-29; trades in holdout: notebook 24, lab 24. Small differences can arise if Yahoo revised adjusted prices between downloads.

Full available history (after 273-bar warm-up)

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	In mkt	Entries	Win%	Turnover/ yr
Buy-and-hold	10.9%	18.7%	0.51	0.73	-55.2%	100.0%	1	100.0 %	0.0
F01_01 SMA 20/50 crossover (original)	6.6%	12.0%	0.38	0.52	-31.2%	67.9%	85	49.4%	5.2
F01_02 SMA 10/30 crossover (original)	6.1%	11.5%	0.35	0.48	-29.7%	65.8%	136	44.1%	8.3
F02_01 Asymmetric price/SMA: exitSMA10 (original)	2.2%	11.7%	0.03	0.04	-55.3%	66.6%	632	36.2%	38.8
F02_07 Two-stage 0/50/100 SMA10/20 (original)	1.0%	10.8%	-0.08	-0.11	-51.8%	70.4%	548	30.1%	36.9
F05_09 10-day breakout + 2 ATR stop (original)	3.7%	10.2%	0.16	0.22	-40.1%	56.4%	208	43.8%	12.7
F03_10 5-day momentum + SMA100 regime (original)	8.8%	12.3%	0.55	0.75	-25.7%	75.1%	50	44.0%	3.0
F02_10 Exit on SMA10/30 cross, re-enter on SMA20/50 + SMA10/30	5.9%	10.3%	0.36	0.50	-20.9%	56.3%	117	44.4%	7.1
F10_10 SMA10/30 with SMA200 regime scaling	6.0%	10.3%	0.38	0.51	-23.2%	65.8%	136	43.4%	8.4

Common period 2011-2026

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	In mkt	Entries	Win%	Turnover/ yr
Buy-and-hold	14.1%	17.0%	0.77	1.08	-33.7%	100.0%	1	100.0 %	0.0
F01_01 SMA 20/50 crossover (original)	7.6%	11.5%	0.56	0.75	-27.3%	72.9%	39	61.5%	4.8
F01_02 SMA 10/30 crossover (original)	7.1%	10.9%	0.54	0.74	-17.2%	69.3%	62	45.2%	7.7

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	In mkt	Entries	Win%	Turnover/yr
F02_01 Asymmetric price/SMA: exitSMA10 (original)	4.1%	10.7%	0.28	0.38	-25.7%	70.0%	296	42.2%	37.5
F02_07 Two-stage 0/50/100 SMA10/20 (original)	2.6%	9.8%	0.14	0.19	-24.9%	73.7%	261	34.1%	36.1
F05_09 10-day breakout + 2 ATR stop (original)	5.4%	9.2%	0.45	0.61	-21.6%	58.3%	104	48.1%	13.1
F03_10 5-day momentum + SMA100 regime (original)	8.6%	12.1%	0.61	0.83	-25.7%	82.6%	22	45.5%	2.7
F02_10 Exit on SMA10/30 cross, re-enter on SMA20/50 + SMA10/30	5.6%	9.7%	0.45	0.61	-17.5%	60.9%	55	40.0%	6.9
F10_10 SMA10/30 with SMA200 regime scaling	6.4%	9.9%	0.52	0.70	-16.8%	69.3%	62	45.2%	8.0

Prior notebook development period 2010-01-04 to 2021-09-28 (exploratory)

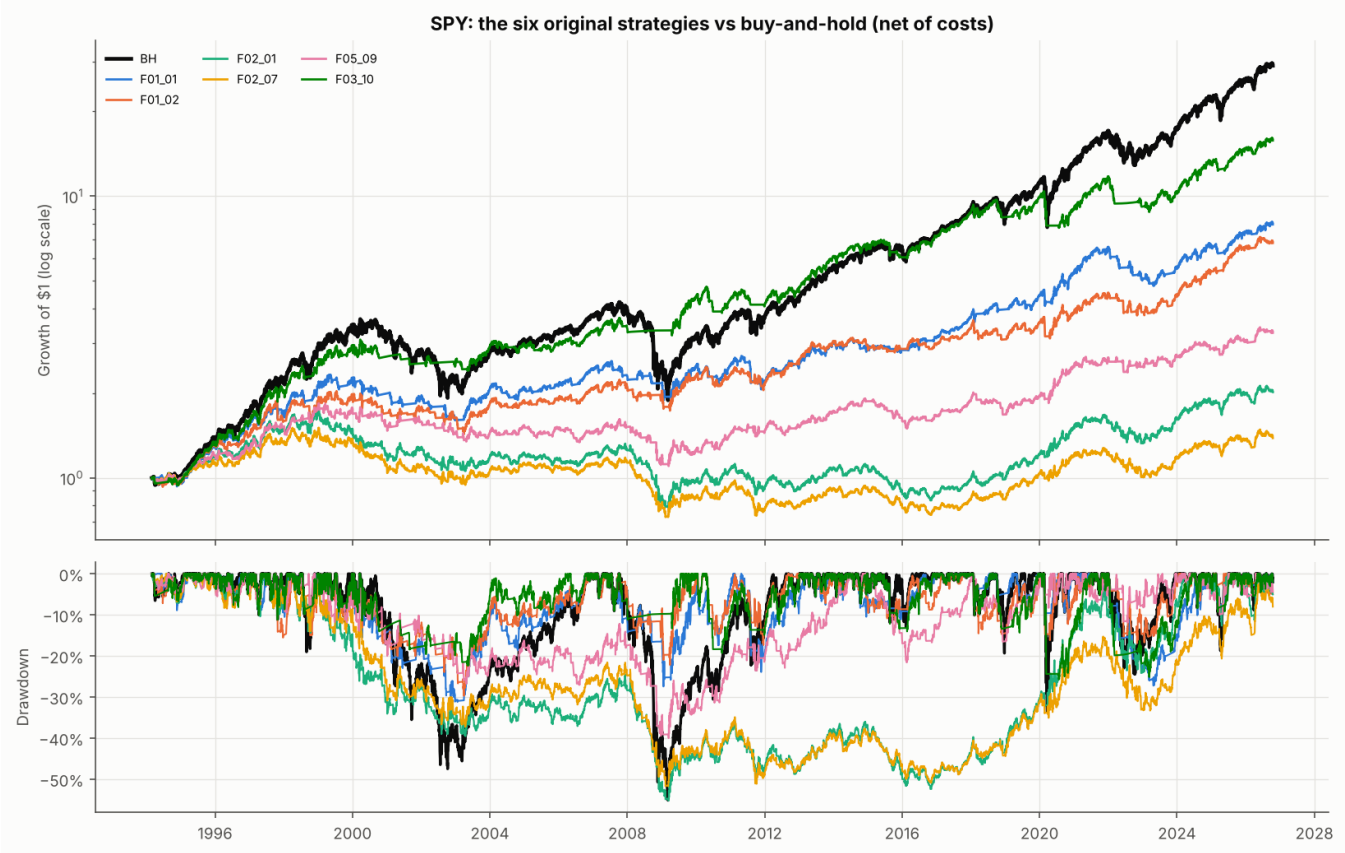
Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	In mkt	Entries	Win%	Turnover/yr
Buy-and-hold	14.5%	17.0%	0.85	1.18	-33.7%	100.0%	1	100.0%	0.0
F01_01 SMA 20/50 crossover (original)	8.4%	11.7%	0.70	0.95	-23.9%	74.1%	30	60.0%	4.9
F01_02 SMA 10/30 crossover (original)	5.7%	10.9%	0.51	0.69	-17.4%	70.6%	47	44.7%	7.9
F02_01 Asymmetric price/SMA: exitSMA10 (original)	4.3%	10.5%	0.40	0.55	-25.7%	71.2%	214	41.6%	36.3
F02_07 Two-stage 0/50/100 SMA10/20 (original)	3.1%	9.7%	0.31	0.42	-24.9%	74.6%	189	34.9%	35.7
F05_09 10-day breakout + 2 ATR stop (original)	6.2%	9.3%	0.64	0.88	-21.6%	60.5%	74	48.6%	12.5
F03_10 5-day momentum + SMA100 regime (original)	7.9%	12.4%	0.63	0.85	-25.7%	83.4%	17	35.3%	2.7
F02_10 Exit on SMA10/30 cross, re-enter on SMA20/50 + SMA10/30	4.2%	9.9%	0.42	0.56	-17.5%	62.1%	41	41.5%	6.9
F10_10 SMA10/30 with SMA200 regime scaling	4.7%	10.1%	0.45	0.60	-16.8%	70.6%	47	42.6%	8.2

Prior notebook holdout 2021-09-29 to 2026-10-16 (already inspected: exploratory, NOT an untouched test)

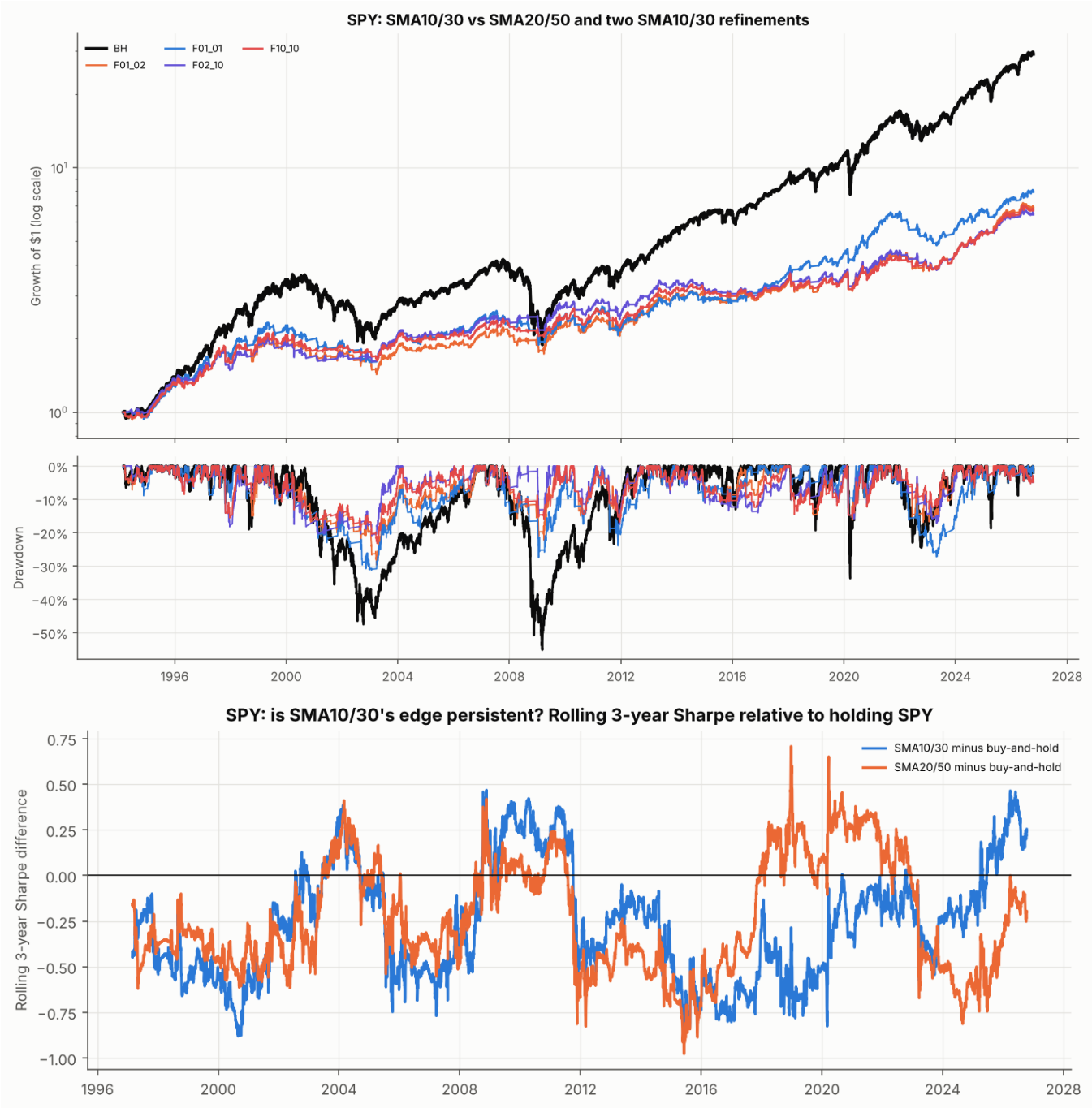
Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	In mkt	Entries	Win%	Turnover/yr
Buy-and-hold	13.4%	17.1%	0.60	0.86	-24.5%	100.0%	1	100.0%	0.0
F01_01 SMA 20/50 crossover (original)	4.7%	11.4%	0.12	0.16	-27.3%	68.8%	13	46.2%	4.8
F01_02 SMA 10/30 crossover (original)	9.4%	10.9%	0.52	0.74	-16.9%	65.6%	20	45.0%	7.7
F02_01 Asymmetric price/SMA: exitSMA10 (original)	4.9%	11.3%	0.14	0.20	-20.7%	67.0%	96	40.6%	38.3
F02_07 Two-stage 0/50/100 SMA10/20 (original)	2.7%	10.2%	-0.07	-0.09	-21.3%	71.5%	86	30.2%	35.7
F05_09 10-day breakout + 2 ATR stop (original)	5.3%	9.3%	0.19	0.27	-11.8%	54.2%	35	45.7%	13.7
F03_10 5-day momentum + SMA100 regime (original)	8.1%	11.6%	0.40	0.56	-25.2%	77.5%	8	62.5%	2.8
F02_10 Exit on SMA10/30 cross, re-enter on SMA20/50 + SMA10/30	7.6%	9.4%	0.42	0.58	-16.9%	56.9%	18	33.3%	7.0

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	In mkt	Entries	Win%	Turnover/ yr
F10_10 SMA10/30 with SMA200 regime scaling	9.7%	9.6%	0.61	0.86	-13.0%	65.6%	20	50.0%	7.9

Equity curves and drawdowns



SMA10/30 focus



Statistical evidence for SPY strategies

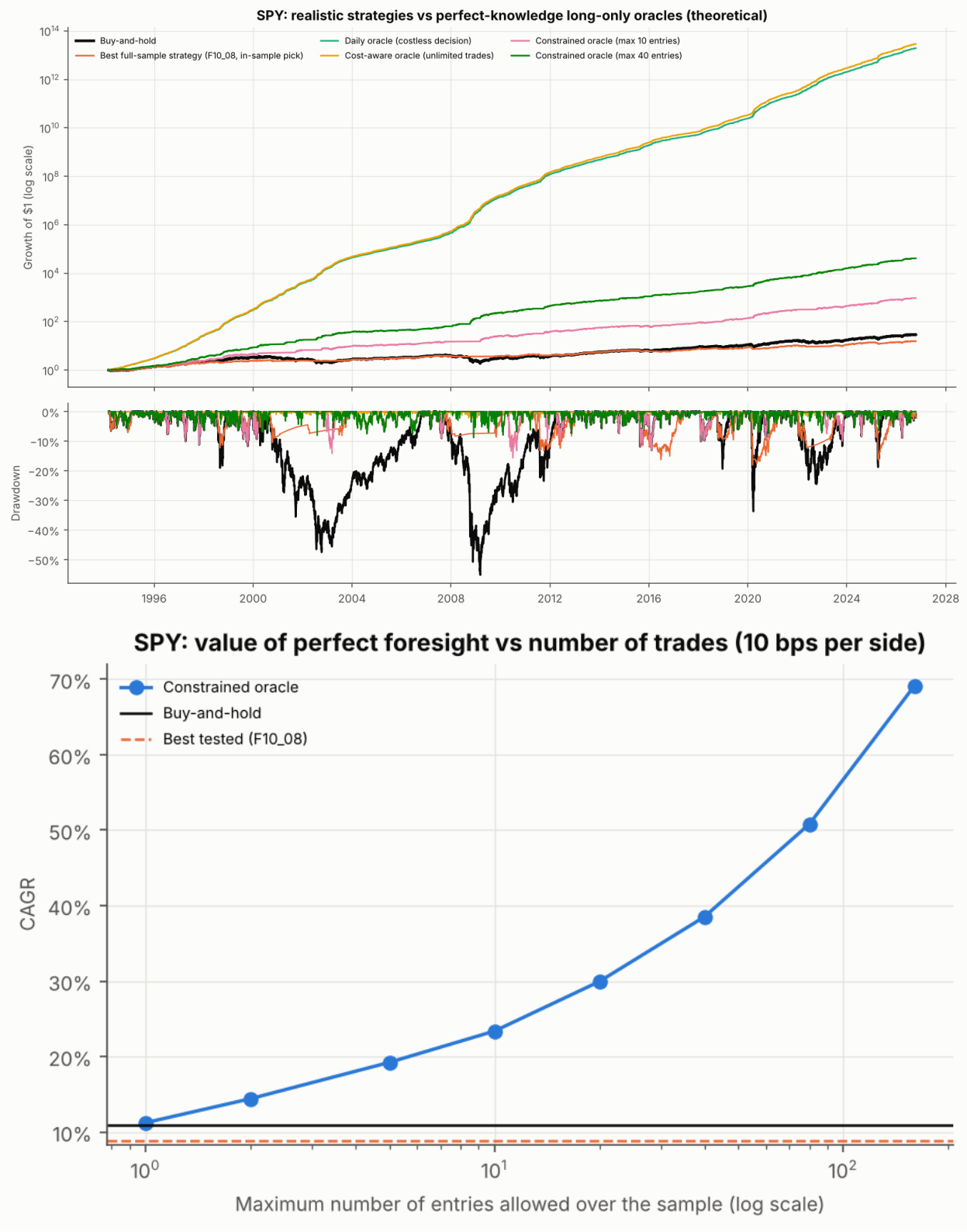
strategy	sharpe	sharpe_ci_lo	sharpe_ci_hi	dsr	p_boot_diff_gt0	q_bh_global	wf_oos_sharpe_diff_mean	wf_share_folds_beat_bh	excess_cagr_vs_vol_matched	dd_improvement_vs_vol_matched	missed_rallies_log	losses_avoided_log
F03_10	0.55	0.28	0.83	0.71	0.94	1.00	-0.38	0.19	0.00	0.15	13.01	12.50
BH	0.51	0.27	0.78	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
F01_01	0.38	0.13	0.66	0.36	0.99	1.00	-0.37	0.30	-0.02	0.09	14.82	13.69
F10_10	0.38	0.10	0.65	0.34	1.00	1.00	-0.46	0.11	-0.02	0.10	17.55	16.36
F02_10	0.36	0.09	0.66	0.32	1.00	1.00	-0.43	0.26	-0.02	0.12	18.94	17.67
F01_02	0.35	0.08	0.63	0.30	1.00	1.00	-0.38	0.19	-0.02	0.06	15.91	14.74

strategy	sharpe	sharpe_ci_lo	sharpe_ci_hi	dsr	p_boot_diff_gt0	q_bh_global	wf_oos_sharpe_diff_mean	wf_share_folds_beat_bh	excess_cagr_vs_vol_matched	dd_improvement_vs_vol_matched	missed_rallies_log	losses_avoided_log
F05_09	0.16	-0.12	0.43	0.05	1.00	1.00	-0.60	0.22	-0.04	-0.07	19.16	17.40
F02_01	0.03	-0.25	0.30	0.01	1.00	1.00	-0.65	0.26	-0.06	-0.16	15.66	14.25
F02_07	-0.08	-0.36	0.20	0.00	1.00	1.00	-0.76	0.22	-0.07	-0.18	16.88	15.04

p_boot: one-sided stationary-bootstrap p-value that the daily return differential vs buy-and-hold is positive; q: Benjamini-Hochberg across all tests. Vol-matched columns compare with a static SPY/cash mix of equal volatility. Missed rallies / losses avoided: sum of log excess returns over cash while out of the market.

H2 across all instruments (Not resolved). Drawdown shallower for 10/30 on 53% of 32 assets (median +0.013, p=0.860); sideways-regime excess return lower for 10/30 on 47% (median +0.186%/yr, p=0.860); Sharpe higher for 10/30 on 44%

Perfect-trader comparison (theoretical)



Ledger reconciliation (equity rebuilt from the trade log vs engine equity)

strategy	max_abs_error	max_rel_error	n_trades
F01_01	3.49e-09	6.06e-15	175

strategy	max_abs_error	max_rel_error	n_trades
F01_02	5.01e-09	8.36e-15	283
F02_01	1.05e-08	8.27e-14	1312
F02_07	7.70e-09	7.24e-14	2034
F05_09	4.37e-09	1.48e-14	429
F03_10	3.49e-09	2.99e-15	99
F02_10	4.66e-09	8.29e-15	241
F10_10	4.77e-09	8.19e-15	345
F10_07	2.10e-09	2.88e-15	1230